



GIRRAWEEEN HIGH SCHOOL
YEAR 12 MATHEMATICS EXTENSION 2
TASK 1 2021
(December 2020)
ANSWERS COVER SHEET

Student Number:

QUESTION	MARK	MEX12-1	MEX12-2	MEX12-3	MEX12-4	MEX12-5	MEX12-6	MEX12-7	MEX12-8
1-5	/5				√				√
6	/17				√				√
7	/10				√				√
8	/6				√				√
9	/15				√				√
10	/14				√				√
11	/10				√				√
TOTALS	/77				/77				/77

Year 12 Mathematics Extension 2 outcomes

A student:

- | | |
|----------------|---|
| MEX12-1 | understands and uses different representations of numbers and functions to model, prove results and find solutions to problems in a variety of contexts |
| MEX12-2 | chooses appropriate strategies to construct arguments and proofs in both practical and abstract settings |
| MEX12-3 | uses vectors to model and solve problems in two and three dimensions |
| MEX12-4 | uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems |
| MEX12-5 | applies techniques of integration to structured and unstructured problems |
| MEX12-6 | uses mechanics to model and solve practical problems |
| MEX12-7 | applies various mathematical techniques and concepts to model and solve structured, unstructured and multi-step problems |
| MEX12-8 | communicates and justifies abstract ideas and relationships using appropriate language, notation and logical argument |



GIRRAWEEEN HIGH SCHOOL

MATHEMATICS EXTENSION 2 2021 HSC Task 1 (December 2020)

Student No: _____

This Booklet contains the answer sheet for Part A and Writing Booklet for Part B.

Part A ANSWER SHEET

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☐	C	☐	D	☐
2.	A	☐	B	☐	C	☐	D	☐
3.	A	☐	B	☐	C	☐	D	☐
4.	A	☐	B	☐	C	☐	D	☐
5.	A	☐	B	☐	C	☐	D	☐

Instructions

- If you need more paper for Part B, please ask your supervisor.
- Write your name on every booklet you use.
- Write on both sides of each sheet of paper.

Total number of booklets used _____.



GIRRAWEEEN HIGH SCHOOL

TASK 1 2021 (*December 2020*)

MATHEMATICS

EXTENSION 2

Complex Numbers

Time allowed – 90 Minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be marked clearly Question 6, Question 7 etc. Each question is to be returned on a separate page in your answer booklet.
- You may ask for spare answer booklets if you need them.
- For Multiple choice: Fill in the circle corresponding to the correct answer on the multiple choice answer sheet in your answer booklet.
- For Question 9(a) please draw your diagrams on the page of number planes provided in your answer booklet.
- The notation *cis* θ is acceptable in this examination.

Multiple Choice begins on the following page

Question 1

$$i^{2021} =$$

- (A) i (B) -1 (C) $-i$ (D) 1

Question 2

$$2e^{\frac{5\pi i}{6}} =$$

- (A) $\sqrt{3} + i$ (B) $\sqrt{3} - i$ (C) $-\sqrt{3} + i$ (D) $-\sqrt{3} - i$

Question 3

$$\frac{z}{\bar{z}} =$$

- (A) $\frac{z^2}{|z|^2}$ (B) $\frac{(\bar{z})^2}{|z|^2}$ (C) $\frac{z\bar{z}}{|z|^2}$ (D) $\frac{1}{z}$

Question 4

In Modulus/argument form to the nearest degree, $-2 + 7i =$

- (A) $\sqrt{53} \text{ cis } 16^\circ$ (B) $\sqrt{53} \text{ cis } 74^\circ$ (C) $\sqrt{53} \text{ cis } 106^\circ$ (D) $\sqrt{53} \text{ cis } 286^\circ$

Question 5

$$\frac{6+8i}{2+i} =$$

- (A) $3 + 8i$ (B) $\frac{4+22i}{5}$ (C) $\frac{20+10i}{3}$ (D) $4 + 2i$

Question 6 (17 Marks – show all of your working in your answer booklet)**Marks**

- (a) (i) Find $\frac{-1+i}{1+i\sqrt{3}}$ in Cartesian form.

2

- (ii) Express $-1 + i$ and $1 + i\sqrt{3}$ in modulus/argument form.

4

- (iii) Hence find an exact value for $\sin \frac{5\pi}{12}$.

2

QUESTION 6 CONTINUES ON THE FOLLOWING PAGE

Question 6 (continued)**Marks**

- (b) (i) By letting $(x + iy)^2 = -16 + 30i$, find $\sqrt{-16 + 30i}$ in Cartesian form.

4

- (ii) Hence solve the equation $z^2 - (1 + i)z + (4 - 7i) = 0$.
Give your answer in Cartesian form.

3

- (c) Find the six sixth roots of $32 - 32i\sqrt{3}$. Leave your answers in modulus/argument form.

2**Question 7 (10 Marks – show all of your working in your answer booklet)**

- (a) Sketch these curves on separate Argand diagrams:

(i) $|z - 1 - 2i| = \sqrt{5}$

4

(ii) $\text{Arg}(z - 2 - i) = \frac{3\pi}{4}$

2

- (b) Sketch the region on the complex plane where $z\bar{z} \leq 4$
and $-\frac{\pi}{4} < \text{Arg } z \leq \frac{\pi}{2}$

4**Question 8 (6 Marks – show all of your working in your answer booklet)**

If α, β and γ are the roots of the equation $2z^3 - 3z^2 + 8z + 5 = 0$,

- (a) Without finding the roots, find the value of $\alpha^2 + \beta^2 + \gamma^2$.

2

- (b) HENCE (using your answer to (a), explain why
 $2z^3 - 3z^2 + 8z + 5 = 0$ has exactly one real root.

2

- (c) If one of the roots of $2z^3 - 3z^2 + 8z + 5 = 0$ is $z = 1 + 2i$,
find, with reasons, the other two roots.

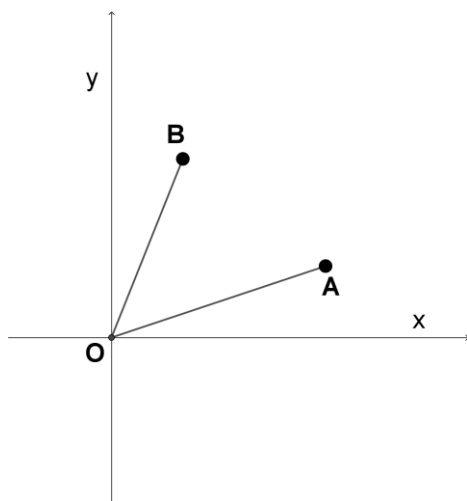
2

EXAMINATION CONTINUES ON THE FOLLOWING PAGE

Question 9 (15 Marks – show all of your working in your answer booklet)

MARKS

- (a) In the diagram below, $\overrightarrow{OA} = z$ and $\overrightarrow{OB} = w$.



Draw and label on the separate Argand diagrams provided in your answer booklet (note that all of these are to be drawn as vectors from O).

6

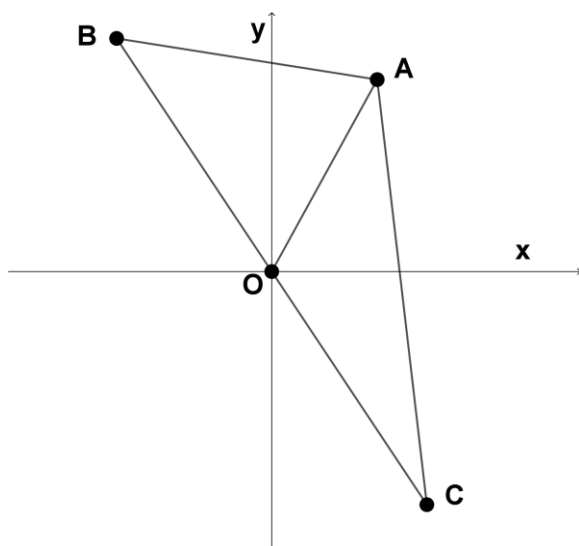
- (i) $\overrightarrow{OC} = z + w$
- (ii) $\overrightarrow{OD} = z - w$
- (iii) $\overrightarrow{OE} = z + 1$
- (iv) $\overrightarrow{OF} = w + i$
- (v) $\overrightarrow{OG} = iw$
- (vi) $\overrightarrow{OH} = z \operatorname{cis}\left(-\frac{\pi}{3}\right)$

QUESTION 9 CONTINUES ON THE FOLLOWING PAGE

Question 9 (continued)

Marks

- (b) In the Argand diagram below, $\overrightarrow{OA} = z_1$ and $\overrightarrow{OB} = z_2$. O is the midpoint of B and C . (see diagram)



Show that $(\overrightarrow{AB})^2 + (\overrightarrow{AC})^2 = 2z_1^2 + 2z_2^2$.

4

- (c) The locus $\text{Arg} \left\{ \frac{z-3}{z+1} \right\} = \frac{\pi}{6}$ represents part of a circle.

5

Draw this circle part on an Argand diagram and find its centre, radius and equation.

Question 10 (14 Marks – show all of your working in your answer booklet)

- (a) Use DeMoivre's Theorem to show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$.

3

- (b) (i) Use Euler's Theorem to show that $e^{ni\theta} + e^{-ni\theta} = 2\cos n\theta$.

2

(ii) HENCE show that $16\cos^5 \theta = \cos 5\theta + 5\cos 3\theta + 10\cos \theta$.

3

(iii) HENCE show that $\cos \frac{\pi}{10}$ is a root of the equation

3

$16x^4 - 20x^2 + 5 = 0$ (you may use your answer to Question 10(a) here)

(iv) HENCE find the exact value of $\cos \frac{\pi}{10}$ (you may leave your answer

3

under a square root sign).

EXAMINATION CONTINUES ON THE FOLLOWING PAGE

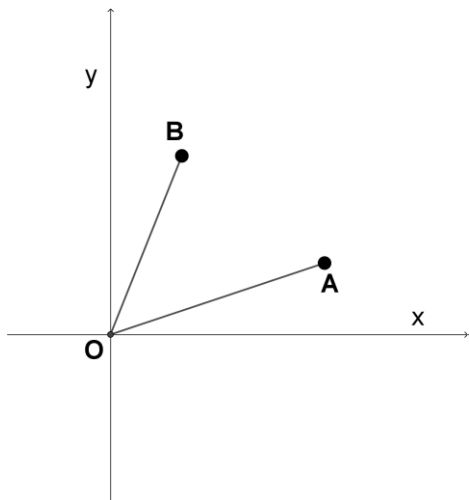
Question11 (10 Marks – show all of your working in your answer booklet) **Marks**

- (a) Solve $z^7 - 1 = 0$. (Leave your answers in $\text{cis } \theta$ form). **1**
- (b) If w is the root of $z^7 - 1 = 0$ with the smallest positive argument, show that w^n is also a root if n is a positive integer. **1**
- (c) Show that $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0$ **1**
- (d) Form the monic quadratic equation with roots $w + w^2 + w^4$ and $w^3 + w^5 + w^6$ **3**
- (e) Hence show that $\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$ **4**

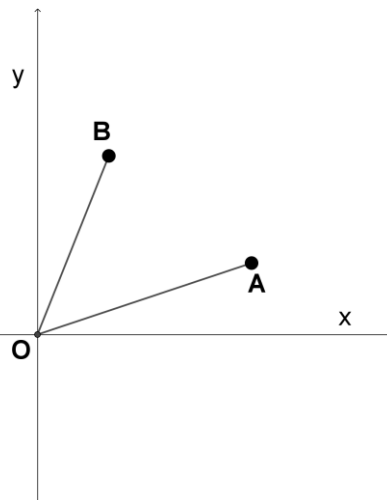
END OF EXAMINATION

Question 9(a) Diagrams

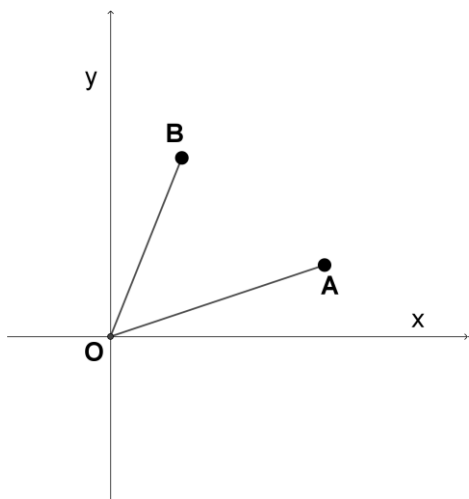
(i)



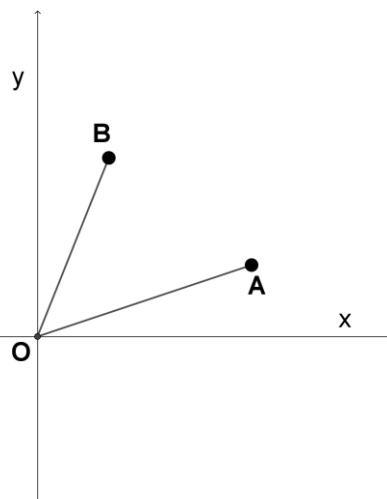
(ii)



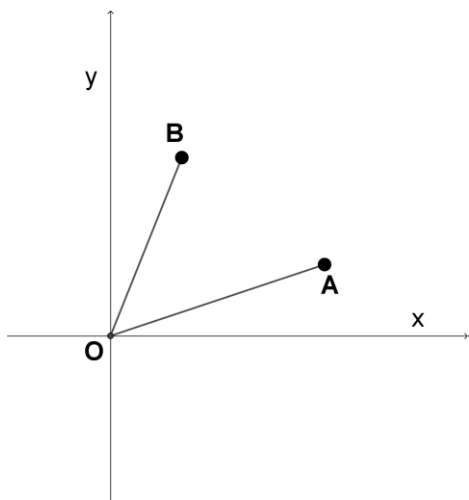
(iii)



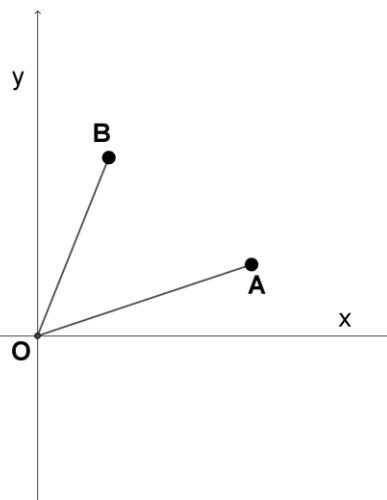
(iv)



(v)



(vi)



Solutions: Ext 2 Task 1 2021 (p.1)
[Dec. 2020]

Complex Numbers
Multiple Choice

(1) A (2) C (3) A (4) C (5) D

$$\begin{aligned} (1) \quad i^{-2021} \\ &= (i^{-2020}) \times i \\ &= 1 \times i \\ &= i \quad \text{(A)} \end{aligned}$$

$$\begin{aligned} (2) \quad 2e^{\frac{5\pi i}{6}} \\ &= 2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}) \\ &= 2\left(-\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) \\ &= -\sqrt{3} + i. \quad \text{(C)} \end{aligned}$$

$$\begin{aligned} (3) \quad \frac{z}{z} \times \frac{z}{z} \\ &= \frac{z^2}{z^2} \\ &= \frac{z}{|z|^2} \quad \text{(A)} \end{aligned}$$

$$\begin{aligned} (4) \quad |-2+7i| \\ &= \sqrt{2^2+7^2} \\ &= \sqrt{53}. \\ \text{Arg}(-2+7i) \\ &= 180^\circ - \tan^{-1}\left(\frac{7}{2}\right) \\ &= 106^\circ. \end{aligned}$$

$$\begin{aligned} \therefore -2+7i &= \sqrt{53} \text{ cis } 106^\circ \\ & \text{(nearest degree)} \quad \text{(C)} \end{aligned}$$

$$\begin{aligned} (5) \quad \frac{6+8i}{2+i} \times \frac{2-i}{2-i} \\ &= \frac{20+10i}{2^2+1^2} \quad \text{(D)} \\ &= 4+2i \end{aligned}$$

$$\begin{aligned} \text{Q. (6) (a) (i)} \quad \frac{-1+i}{1+i\sqrt{3}} \times \frac{(1-i\sqrt{3})}{(1-i\sqrt{3})} \\ &= \frac{-1+i\sqrt{3}+i+\sqrt{3}}{1^2+(\sqrt{3})^2} \quad \underline{2} \\ &= \frac{(\sqrt{3}-1)}{4} + \frac{i(\sqrt{3}+1)}{4} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad -1+i &= \sqrt{2} \text{ cis } \frac{3\pi}{4} \\ 1+i\sqrt{3} &= 2 \text{ cis } \frac{\pi}{3} \quad \underline{4} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \frac{-1+i}{1+i\sqrt{3}} \\ &= \frac{\sqrt{2} \text{ cis } \frac{3\pi}{4}}{2 \text{ cis } \frac{\pi}{3}} \quad \underline{2} \\ &= \frac{1}{\sqrt{2}} \text{ cis } \frac{5\pi}{12} \end{aligned}$$

Equating imaginary parts of (i) & (ii)

$$\begin{aligned} \frac{\sqrt{3}+1}{4} &= \frac{1}{\sqrt{2}} \sin \frac{5\pi}{12} \\ \therefore \sin \frac{5\pi}{12} &= \frac{\sqrt{6}+\sqrt{2}}{4} \quad \underline{2} \end{aligned}$$

Solutions: Complex Nos. Dec. 2020 p(2)

Q.(6)(b)(i) $(x+iy)^2 = -16+30i$, x, y real.

$$(x^2 - y^2) + 2ixy = -16 + 30i.$$

Equating reals,

$$x^2 - y^2 = -16 \quad (1)$$

Equating imaginaries

$$2xy = 30$$

$$y = \frac{15}{x} \quad (2)$$

Sub. (2) in (1)

$$x^2 - \left(\frac{15}{x}\right)^2 = -16$$

$$x^2 - \frac{225}{x^2} = -16$$

$$x^4 + 16x^2 - 225 = 0$$

$$(x^2 + 25)(x^2 - 9) = 0$$

$$x = \pm 3 \text{ as } x \in \mathbb{R}.$$

$$\text{Sub. } x = 3 \text{ in (2) } y = \frac{15}{3} = 5.$$

$$\text{Sub. } x = -3 \text{ in (2) } y = \frac{15}{-3} = -5.$$

$$\sqrt{-16+30i} = \pm (3+5i).$$

4

(ii) $z^2 - (1+i)z + (4-7i) = 0$

$$z = \frac{(1+i) \pm \sqrt{(1+i)^2 - 4 \times 1 \times (4-7i)}}{2 \times 1}$$

$$= \frac{(1+i) \pm \sqrt{-16+30i}}{2}$$

$$z = \frac{1+i + (3+5i)}{2} \quad \text{or} \quad z = \frac{1+i - (3+5i)}{2} \quad \underline{3}$$

$$z = 2+3i \quad \text{or} \quad z = -1-2i.$$

(c) $32 - 32i\sqrt{3} = 64 \operatorname{cis}\left(-\frac{\pi}{3}\right)$

$$\therefore \sqrt[6]{32-32i\sqrt{3}} = 2 \operatorname{cis}\left(\frac{-\pi}{18} + \frac{2k\pi}{6}\right), \quad k=0,1,2,3,4,5. \text{ By De Moivre. } \underline{2}$$

$$= 2 \operatorname{cis}\left(-\frac{\pi}{18}\right), 2 \operatorname{cis}\left(\frac{5\pi}{18}\right), 2 \operatorname{cis}\left(\frac{11\pi}{18}\right), 2 \operatorname{cis}\left(\frac{17\pi}{18}\right), 2 \operatorname{cis}\left(\frac{23\pi}{18}\right), 2 \operatorname{cis}\left(\frac{29\pi}{18}\right)$$

$$\text{if } -\pi < \operatorname{Arg} z \leq \pi = 2 \operatorname{cis}\left(-\frac{\pi}{18}\right), 2 \operatorname{cis}\left(\frac{5\pi}{18}\right), 2 \operatorname{cis}\left(\frac{11\pi}{18}\right), 2 \operatorname{cis}\left(\frac{17\pi}{18}\right), 2 \operatorname{cis}\left(-\frac{13\pi}{18}\right), 2 \operatorname{cis}\left(-\frac{7\pi}{18}\right).$$

Solutions: Complex N^os Dec. 2020 p. (3)

Q. (7) (a) (i) $|z - 1 - 2i| = \sqrt{5}$

→ Circle centre $1 + 2i$, $r = \sqrt{5}$.

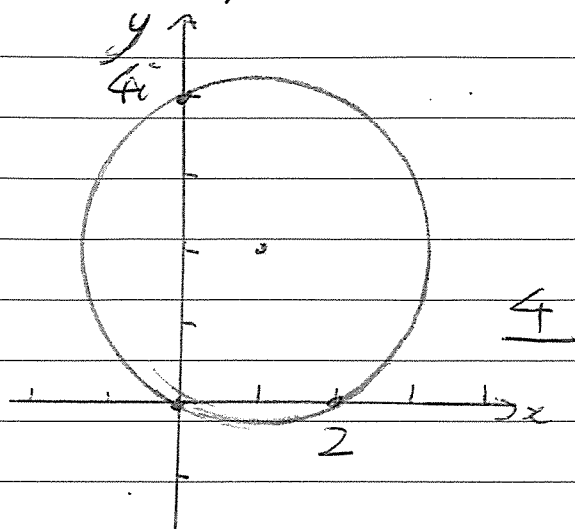
Equation $(x-1)^2 + (y-2)^2 = 5$.

x intercepts: $(x-1)^2 + (0-2)^2 = 5$.

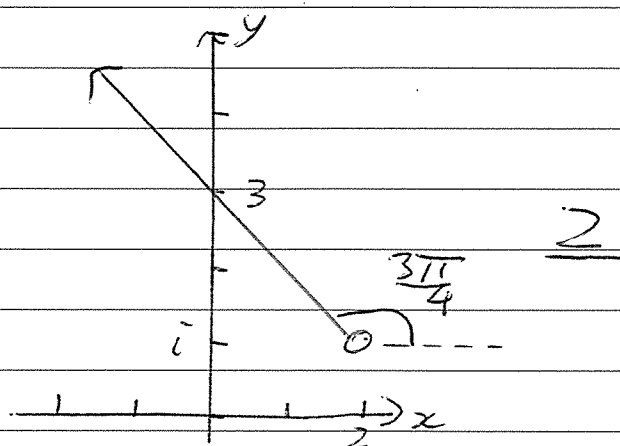
$x = 2$ or $x = 0$

y intercepts: $(0-1)^2 + (y-2)^2 = 5$

$y = 0$ or $4i$



(ii) $\text{Arg}(z - 2 - i) = \frac{3\pi}{4}$



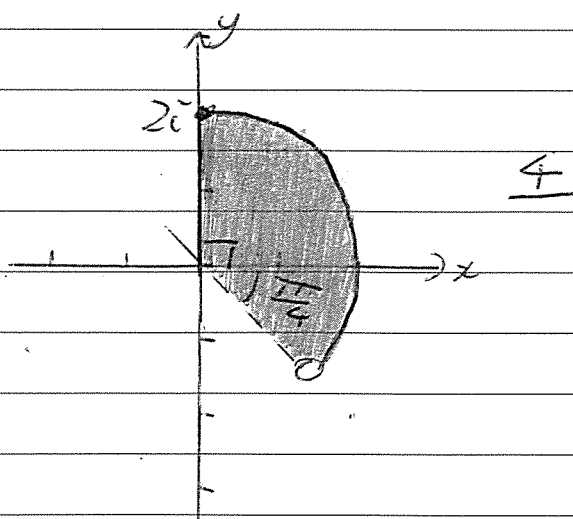
(b) $z\bar{z} \leq 4$

If $z = x + iy$,

$z\bar{z} = (x + iy)(x - iy)$

$= x^2 + y^2$

So $x^2 + y^2 \leq 4$.



$$\begin{aligned}
 \text{Q. (8) (a)} \quad & \alpha^2 + \beta^2 + \gamma^2 \\
 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \quad \left| \begin{array}{l} \text{Note: } \alpha + \beta + \gamma = \frac{3}{2} \\ \alpha\beta + \alpha\gamma + \beta\gamma = \frac{8}{2} = 4. \end{array} \right. \\
 &= \left(\frac{3}{2}\right)^2 - 2 \times 4 \\
 &= \frac{-23}{4}.
 \end{aligned}$$

2

(b) As $\alpha^2 + \beta^2 + \gamma^2 < 0$, $2z^3 - 3z^2 + 8z + 5 = 0$ must have at least 1 non-real root.

As all co-efficients are real, the CONJUGATE of that non-real root must also be a root.

2

$\therefore 2z^3 - 3z^2 + 8z + 5 = 0$ has 3 roots [fundamental theorem of algebra].

$2z^3 - 3z^2 + 8z + 5 = 0$ has 1 real root [2 non real roots].

(c) If $1 + 2i$ is a root, then its conjugate $(1 - 2i)$ is also a root

$$\begin{aligned}
 \therefore \text{As } \alpha + \beta + \gamma &= \frac{3}{2} \\
 (1 + 2i) + (1 - 2i) + \gamma &= \frac{3}{2} \\
 \gamma &= \frac{-1}{2}.
 \end{aligned}$$

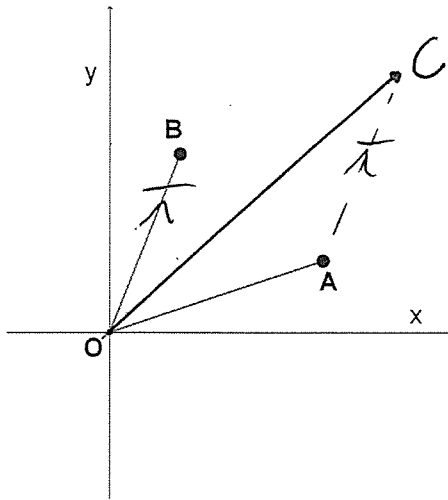
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$\therefore$ Roots of $2z^3 - 3z^2 + 8z + 5 = 0$ are $1 + 2i, 1 - 2i, -\frac{1}{2}$.

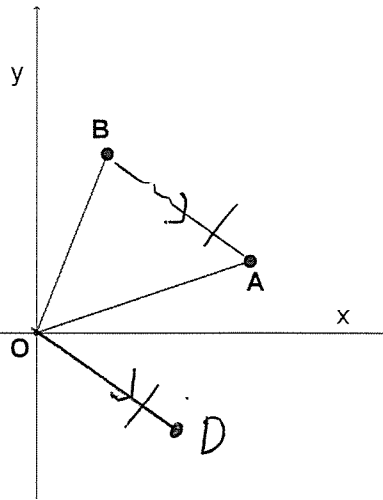
For Q. (9) (a) PTD $\rightarrow$

Question 9(a) Diagrams

(i)



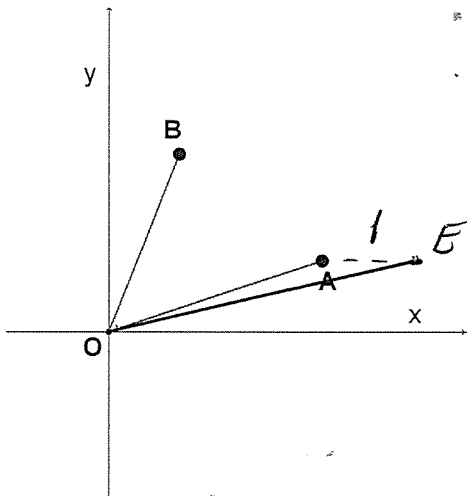
(ii)



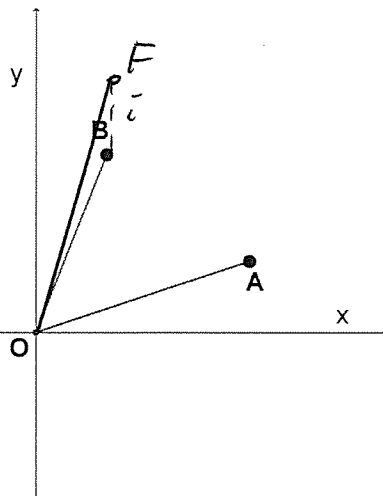
$$\vec{OB} = \vec{BA}$$

6

(iii)



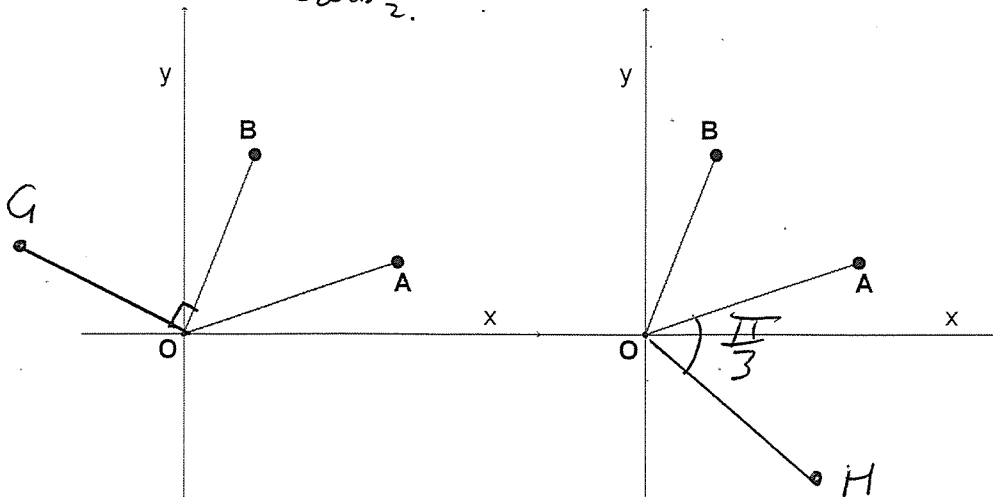
(iv)



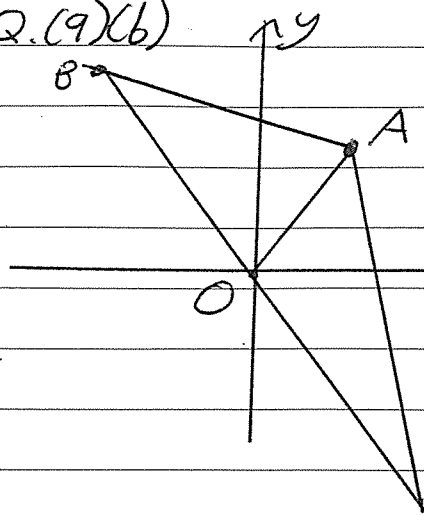
(v)

$$\vec{OG} = i\omega = \omega \cos \frac{\pi}{2}$$

(vi)



Q. (9)(b)



Note: $\vec{OC} = -\vec{OB} = -z_2$

$$\vec{AB} = z_2 - z_1$$

$$\vec{AC} = -z_2 - z_1$$

$$\therefore (\vec{AB})^2 + (\vec{AC})^2$$

$$= (z_2 - z_1)^2 + (-z_2 - z_1)^2$$

$$= z_2^2 - 2z_1z_2 + z_1^2 + z_2^2 + 2z_1z_2 + z_1^2$$

$$= 2z_1^2 + 2z_2^2$$

$$= \text{RHS Q.E.D.}$$

4

1c) Let $\vec{OA} = z$

$$\vec{OB} = 3$$

$$\vec{OC} = -1$$

Let H = circle centre.

$$\angle BHC = \frac{\pi}{3} \quad \begin{aligned} & \text{[at centre} \\ & = 2 \times \text{[at} \\ & \text{circumference} \\ & \text{subtended} \\ & \text{by arc AB)} \end{aligned}$$

As $BH = CH$ [H is circle centre],

$\triangle BHC$ is equilateral.

Also, H is on $x=1$ by symmetry.

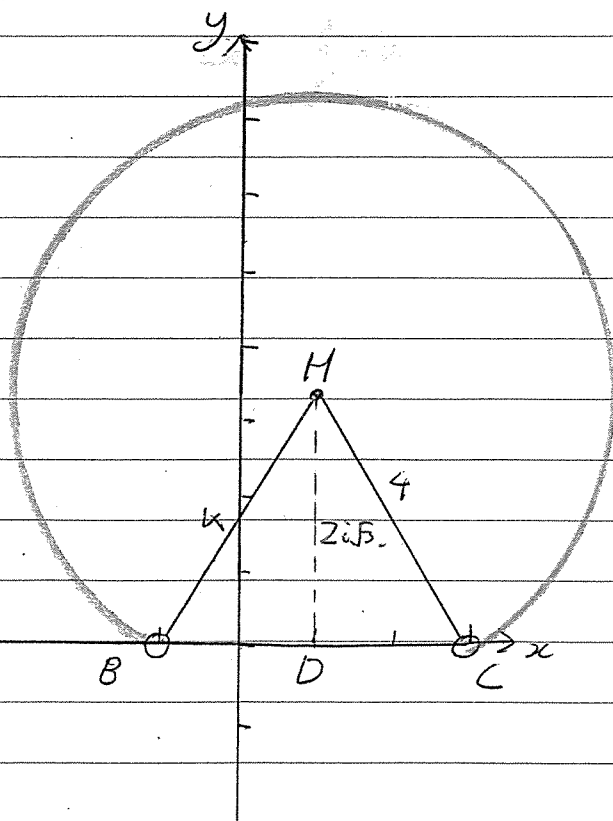
Let $D = 1+0i$ be the midpoint of B & C .

$$BH = CH = 4$$

$$DH = 4 \sin \frac{\pi}{3} = 2\sqrt{3}$$

$$\text{So } \vec{OH} = 1 + 2i\sqrt{3}$$

$$\text{Circle is } (x-1)^2 + (y-2\sqrt{3})^2 = 16, y > 0.$$



Solution: Complex N^os Dec 2020 p. ⑦

(10)(a) By De Moivre's Theorem

$$\begin{aligned} (\cos \theta + i \sin \theta)^3 &= \cos 3\theta + i \sin 3\theta \quad \underline{3} \\ \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta &= \cos 3\theta + i \sin 3\theta \end{aligned}$$

Equating real parts,

$$\begin{aligned} \cos 3\theta &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\ &= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta \\ \cos 3\theta &= 4 \cos^3 \theta - 3 \cos \theta. \end{aligned}$$

(b)(i) As $e^{ni\theta} = \cos n\theta + i \sin n\theta$ (1) + [Euler]

$$\begin{aligned} \& e^{-ni\theta} = \cos n\theta - i \sin n\theta \quad (2) \quad \underline{2} \\ e^{ni\theta} + e^{-ni\theta} &= 2 \cos n\theta \end{aligned}$$

$$\begin{aligned} \text{(ii)} (e^{i\theta} + e^{-i\theta})^5 &= e^{5i\theta} + 5e^{3i\theta} + 10e^{i\theta} + 10e^{-i\theta} + 5e^{-3i\theta} + e^{-5i\theta} \\ &= (e^{5i\theta} + e^{-5i\theta}) + 5(e^{3i\theta} + e^{-3i\theta}) + 10(e^{i\theta} + e^{-i\theta}) \end{aligned}$$

Using Euler's theorem,

$$\begin{aligned} [2 \cos \theta]^5 &= 2 \cos 5\theta + 5 \times 2 \cos 3\theta + 10 \times 2 \cos \theta \quad \underline{3} \\ 32 \cos^5 \theta &= 2 \cos 5\theta + 10 \cos 3\theta + 20 \cos \theta \\ &\div 2 \end{aligned}$$

$$16 \cos^5 \theta = \cos 5\theta + 5 \cos 3\theta + 10 \cos \theta.$$

(iii) If $\theta = \frac{\pi}{10}$, $5\theta = \frac{\pi}{2}$ & $\cos 5\theta = \cos \frac{\pi}{2} = 0$

$$\& 3\theta = \frac{3\pi}{10} \rightarrow \cos \frac{3\pi}{10} = 4 \cos^3 \frac{\pi}{10} - 3 \cos \frac{\pi}{10} \quad \text{(from (a))} \quad \underline{3}$$

$$\text{Sub. } \cos 5\theta = 0 \& \cos \frac{3\pi}{10} = 4 \cos^3 \frac{\pi}{10} - 3 \cos \frac{\pi}{10} \text{ in (b)(ii)}$$

$$16 \cos^5 \frac{\pi}{10} = 0 + 5 \left(4 \cos^3 \frac{\pi}{10} - 3 \cos \frac{\pi}{10} \right) + 10 \cos \frac{\pi}{10}$$

$$16 \cos^5 \frac{\pi}{10} = 20 \cos^3 \frac{\pi}{10} - 5 \cos \frac{\pi}{10} \quad \div \cos \frac{\pi}{10}, \text{ as } \cos \frac{\pi}{10} \neq 0$$

$$16 \cos^4 \frac{\pi}{10} = 20 \cos^2 \frac{\pi}{10} - 5$$

$$16 \cos^4 \frac{\pi}{10} - 20 \cos^2 \frac{\pi}{10} + 5 = 0$$

$$\therefore \cos \frac{\pi}{10} \text{ is a root of } 16x^4 - 20x^2 + 5 = 0$$

PTO $\rightarrow$

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Q.40)(b)(iv) $\cos \frac{\pi}{10}$ is a solution of $16x^4 - 20x^2 + 5 = 0$

By quadratic formula, $x^2 = \frac{20 \pm \sqrt{(-20)^2 - 4 \times 16 \times 5}}{2 \times 16}$

$$x^2 = \frac{5 \pm \sqrt{5}}{8}$$

Note: As $\frac{\pi}{10}$ is in Q_1 , $\cos \frac{\pi}{10}$ will be positive. 3

Also, $\frac{5 - \sqrt{5}}{8} < \frac{1}{2}$ so $\sqrt{\frac{5 - \sqrt{5}}{8}} < \frac{1}{\sqrt{2}}$

which means $\cos^{-1}\left(\sqrt{\frac{5 - \sqrt{5}}{8}}\right) > \frac{\pi}{4}$ & $\neq \frac{\pi}{10}$

So $\cos \frac{\pi}{10} = \sqrt{\frac{5 + \sqrt{5}}{8}}$

Q.41)(a) $z^7 - 1 = 0$

$z = \text{cis } \frac{2k\pi}{7}$, $k = 0, 1, 2, 3, 4, 5, 6$. 1

(b) $w = \text{cis } \frac{2\pi}{7}$

$w^n = \text{cis } \frac{2n\pi}{7}$ [de Moivre]

$(w^n)^7 - 1 = \text{cis } 2n\pi - 1$

$= 1 - 1$

$= 0$

w^n is a solution. 1

(c) Roots of $z^n - 1 = 0 = 1, \text{cis } \frac{2\pi}{7}, \text{cis } \frac{4\pi}{7}, \text{cis } \frac{6\pi}{7}, \text{cis } \frac{8\pi}{7}, \text{cis } \frac{10\pi}{7}, \text{cis } \frac{12\pi}{7}$
 $= 1, w, w^2, w^3, w^4, w^5, w^6$

By sum of roots of $z^n - 1 = 0$, $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0$ 1

(d) Sum of roots: $= w + w^2 + w^4 + w^3 + w^5 + w^6 = -1$. (using (c)).

Product of roots $= (w + w^2 + w^4)(w^3 + w^5 + w^6)$

$= w^4 + w^6 + w^7 + w^5 + w^7 + w^8 + w^7 + w^9 + w^{10}$

$= w^4 + w^6 + 1 + w^5 + 1 + w + 1 + w^2 + w^3$ 3

$= (1 + w + w^2 + w^3 + w^4 + w^5 + w^6) + 2$

$= 2$ [as $1 + w + \dots + w^6 = 0$ from (c)].

$\therefore$ Monic quadratic equation is $z^2 + z + 2 = 0$.

Solutions: Complex N^{os} Dec - 2020 p. 9.

Q. (11)(e) if $w = \text{cis } \frac{2\pi}{7}$,

$$w + w^2 + w^4$$

$$= \left(\cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} \right) + \left(\cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} \right) + \left(\cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7} \right)$$

$$= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} - \cos \frac{3\pi}{7} + i \sin \frac{3\pi}{7} - \cos \frac{\pi}{7} - i \sin \frac{\pi}{7}$$

$$\text{Imaginary part of } w + w^2 + w^4 = \left[\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} \right]$$

$$\text{Roots of } z^2 + z + 2 = 0$$

are z

$$= \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times 2}}{2 \times 1}$$

$$= \frac{-1 \pm i\sqrt{7}}{2}$$

4

Equating imaginary part of $w + w^2 + w^4$ with solution,

$$\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} = \pm \frac{\sqrt{7}}{2}$$

As $\sin \frac{2\pi}{7} \& \sin \frac{3\pi}{7} > \sin \frac{\pi}{7}$ (all in Q_1 & $\frac{2\pi}{7}, \frac{3\pi}{7}$ close to $\frac{\pi}{2}$)

$$\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} > 0$$

$$\sin \frac{2\pi}{7} + \sin \frac{3\pi}{7} - \sin \frac{\pi}{7} = \frac{\sqrt{7}}{2}$$